**C** For each question, draw a triangle ABC and label it with the given information. Then calculate the length of side $a$ correct to 1 decimal place.

1 A = 58°, B = 73° and $b$ = 27 cm
2 A = 43°, B = 110° and $b$ = 38 cm
3 A = 73°, $b$ = 18 cm and $c$ = 14 cm
4 A = 115°, $b$ = 26 cm and $c$ = 22 cm

Find the size of angle A in degrees and minutes.

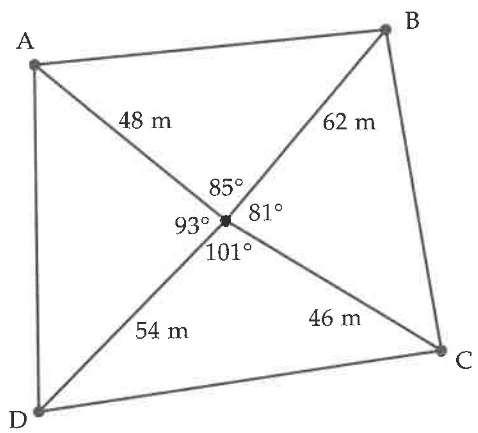
5 B = 65°, $a$ = 18 cm and $b$ = 23 cm
6 C = 37°, $a$ = 46 cm and $b$ = 34 cm
7 $a$ = 15 cm, $b$ = 20 cm and $c$ = 26 cm
8 $a$ = 38 cm, $b$ = 28 cm and $c$ = 16 cm

9 The two equal sides of an isosceles triangle are 165 mm long and the two equal angles are 42°. Find the length of the third side of the triangle, to the nearest millimetre.

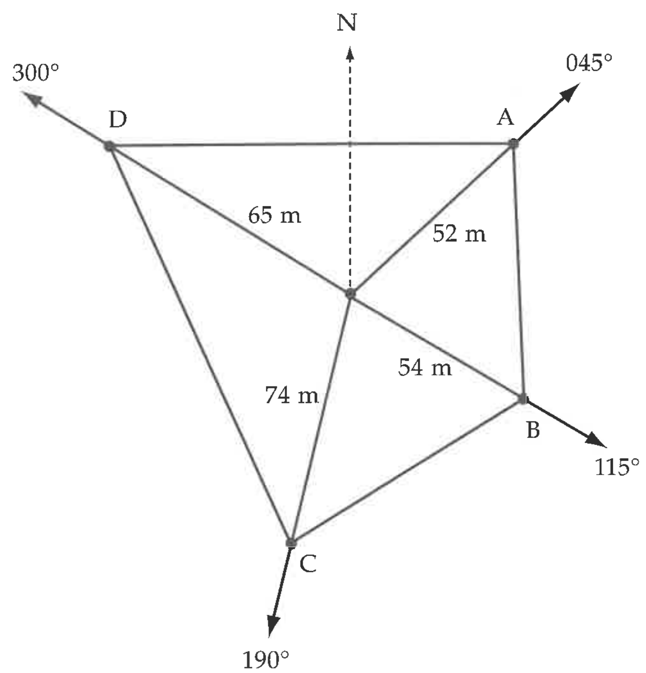
10 Find the size of the largest angle in a triangle with sides measuring 56 mm, 64 mm and 75 mm. (Hint: The largest angle is opposite the longest side.)

11 A balloon is flying in a straight line at an altitude of 600 metres. Matt is standing on the ground. He measures the angle of elevation of the balloon to be 85°. Three minutes later, Matt observed the balloon at an angle of elevation of 20°. How far has the balloon travelled in this time?

12 The piece of land represented in this diagram is based on a radial survey. Calculate its perimeter and area. Round your final answers to the nearest whole number.




13 A radial survey was conducted on a piece of land. A compass was used to determine the directions of the corner posts. The diagram shows all the measurements from the survey. Calculate the perimeter and the area of this piece of land. Round your final answers to the nearest whole number.









### 1 Review: trigonometry of right-angled triangles

**A** 1 17.5 cm 2 24.7 cm 3 10.9 cm 4 18.5 cm 5 49.3 cm 6 7.6 cm 7 32° 8 56° 9 67° 10 55° 11 66° 12 19°

**B** 1 17.4 cm 2 39.7 cm 3 26.1 cm 4 2.1 m 5 5.6 m 6 2.2 m 7 19°11 8 60°17′ 9 36°20′ 10 36°52′ 11 42°34′ 12 56°50′

**C** 1 2.4 m 2 89.5 m 3 117 mm 4 9.2 m 5 5° 6 33.0 m 7 71°47′ 8 112 m **Riddle:** Frostbite

### 2 Exact ratios for 30°, 45° and 60°

**A** 1 1 2 0.5 3 0.5 4 0.707 5 0.707 6 1.732 7 0.866 8 0.866 9 45° 10 45° 11 45° 12 60° 13 60° 14 30° 15 30° 16 45°

**B** 1 $\frac{\sqrt{3}}{2}$ 2 $\frac{1}{2}$ 3 $\sqrt{3}$ 4 1 5 $\frac{\sqrt{3}}{2}$ 6 $\frac{1}{\sqrt{3}}$ 7 $\frac{1}{2}$ 8 $\frac{1}{\sqrt{2}}$

**C** 1 30° 2 60° 3 45° 4 45° 5 60° 6 30° 7 30° 8 60° **Riddle:** Square roots

### 3 Ratios for angles greater than 90°

**A** 1 0.466 2 0.602 3 0.966 4 0.900 5 0.961 6 2.145 7 0.883 8 0.829 9 2.747 10 0.375 11 0.675 12 0.695 13 0.602 14 0.276 15 8.144 16 0.799

**B** 1 0.819 2 −0.259 3 −1.732 4 −0.906 5 0.342 6 −1.192 7 −0.174 8 0.996 9 −0.669 10 0.906 11 −3.732 12 0.469 13 0.643 14 −0.588 15 −0.344 16 −0.174 17 −0.445 18 0.996 19 −0.829 20 0.956 21 0.682 22 −0.951 23 −1.664 24 −0.052 25 0.709 26 −4.247 27 −0.824 28 0.489 29 −0.240 30 0.989 31 −0.273 32 −0.333 33 0.792 34 −0.589 35 −0.511 36 0.977 37 −3.442 38 0.903 39 −0.994 40 −3.785

**C** 1 −0.342 2 −0.966 3 5.671 4 0.574 5 −0.707 6 −1.111 7 −0.866 8 −0.174 9 0.342 10 −0.574 11 1.192 12 −0.5 13 −0.966 14 4.011 15 0.966 16 0.466 17 0.208 18 −0.292 19 −0.407 20 0.777 21 −0.194 22 −0.559 23 −0.978 24 −1.483 25 0.906 26 −0.602 27 −0.900 28 0.276 29 −1.963 30 −0.940 31 0.602 32 −0.990 33 146° 34 134° 35 161° 147° 36 153° 37 104° 38 130° 39 112° 40 136° 41 147° 42 140° 43 129° 44 122° 45 113° 46 115° 47 151° 48 123° 49 112° 50 142° 51 101° 52 111° 53 143° 54 108° 55 118° 56 116° 57 155° 58 155° 59 156° 60 100° 61 125° 62 106° 63 168° 64 144°

### 4 The sine rule: finding the length of a side

**A** 1 0.83 2 0.98 3 0.75 4 0.90 5 0.97 6 0.78 7 0.89 8 0.64 9 6.57 10 3.01 11 27.93 12 3.64 13 83.69 14 4.70 15 41.06 16 3.62

**B** 1 31.9 cm 2 32.4 cm 3 19.7 cm 4 51.2 cm 5 28.1 cm 6 47.5 cm 7 32.1 cm 8 32.6 cm 9 46.8 cm 10 40.6 cm 11 20.9 cm 12 94.1 cm

**C** 1 58.3 cm 2 34.8 cm 3 49.1 cm 4 76.2 cm 5 37.5 cm 6 39.5 cm 7 3.9 m 8 14.9 cm 9 5.5 m 10 7.6 m 11 7.2 m 12 6.7 m 13 22.8 m 14 9.6 m 15 14.1 m

### 5 The sine rule: finding the size of an angle

**A** 1 66° 2 26° 3 49° 4 13° 5 35° 6 11° 7 44° 8 76° 9 125° 10 95° 11 140° 12 155° 13 108° 14 164° 15 161° 16 128°

**B** 1 75° 2 47° 3 61° 4 55° 5 70° 6 55° 7 120° 8 112° 9 110° 10 141° 11 96° 12 132°

**C** 1 48°55′ 2 46°53′ 3 51°37′ 4 14°59′ 5 53°11′ 6 100°17′ 7 37°56′ 8 37°51′ 9 63°36′ 10 33°34′ 11 92°56′ 12 57°37′

### 6 The sine rule: mixed questions

**A** 1 25.7 cm 2 18.8 cm 3 20.4 cm 4 18.8 cm 5 30.8 cm 6 26.6 cm 7 68° 8 60° 9 75° 10 69° 11 133° 12 75°

**B** 1 1.6 m 2 2.8 m 3 3.3 m 4 2.9 m 5 2.5 m 6 3.6 m 7 47°19′ 8 25°26′ 9 75°42′ 10 61°19′ 11 134°46′ 12 101°51′

**C** 1 11.7 cm 2 18.8 cm 3 23.4 cm 4 30.7 cm 5 41°26′ 6 69°6′ 7 41°47′ 8 57°20′ 9 426 m 10 19°

### 7 The cosine rule: finding the length of a side

**A** 1 0.14 2 0.89 3 0.96 4 0.92 5 −0.31 6 −0.86 7 −0.53 8 −0.95 9 13.52 10 −4.08 11 27.89 12 −19.74 13 78.16 14 −14.95 15 −60.53 16 72.89

**B** 1 14.7 cm 2 29.7 cm 3 3.4 cm 4 69.5 cm 5 14.0 cm 6 9.4 cm 7 56.1 cm 8 50.8 cm 9 8.6 cm 10 21.5 cm 11 25.5 cm 12 54.4 cm

**C** 1 7.5 cm 2 48.6 cm 3 12.2 cm 4 36.5 cm 5 8.4 m 6 4.8 m 7 16.7 m 8 30.2 m 9 145.2 m 10 8.8 m 11 24.6 m 12 3.3 m 13 2.4 cm 14 138.9 cm 15 32.3 cm

### 8 The cosine rule: finding the size of an angle

**A** 1 50° 2 61° 3 83° 4 75° 5 68° 6 85° 7 24° 8 36° 9 156° 10 132° 11 139° 12 174° 13 122° 14 92° 15 113° 16 121°

**B** 1 76° 2 79° 3 23° 4 85° 5 22° 6 87° 7 111° 8 128° 9 130° 10 121° 11 115° 12 104°